

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

B.Tech (Electronics and Telecommunication Engineering) SEMESTER - 3 Summer 2025 (Supply.)

Course :Bachelor of Technology (Electronics and Telecommunication Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: BTES304 - ELECTRICAL MACHINES AND INSTRUMENTS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

- Q1. Objective type questions. (Compulsory Question) 12
1. The main function of the commutator in a DC machine is to:
a) Reduce armature reaction b) Convert AC to DC
c) Reverse the direction of current in armature conductors d) Increase efficiency
 2. In a DC motor, the back emf:
a) Increases the armature current b) Helps in starting
c) Opposes the applied voltage d) Acts as a braking force
 3. A four-point starter is used for:
a) Series motors b) Compound motors c) Shunt motors d) All of the above
 4. In an induction motor, the rotor never reaches synchronous speed because:
a) Of high rotor resistance b) Back emf is produced
c) Torque becomes zero d) Core losses are high
 5. Which starter is suitable for a 3-phase squirrel cage induction motor?
a) DOL starter b) Star-delta starter
c) Auto-transformer starter d) All of the above
 6. The device used to improve the power factor using a synchronous motor is:
a) Induction regulator b) V-curve controller
c) Synchronous condenser d) Capacitor bank
 7. A stepper motor operates based on:
a) Electromagnetic induction b) Rotor inertia
c) Discrete steps d) Linear motion
 8. A servo motor is mainly used in:
a) Welding b) Speed control systems c) Drilling d) Transformers
 9. A strain gauge is used to measure:
a) Pressure b) Force c) Temperature d) Displacement
 10. Which of the following transducers is sensitive to light?
a) Strain gauge b) LVDT c) Photodiode d) Thermocouple
 11. Vibration in machines is measured using:
a) Strain gauge b) Tachometer c) Accelerometer d) LVDT

- 12 Smoke detectors work on the principle of:
a) Thermocouple b) Radiation c) Optical sensing d) Induction
- Q2. Solve the following.
- A) What are the different methods of speed control of DC motors? 6
 - B) Explain the necessity of a starter in DC motors. Describe the working of a three-point starter. 6
- Q3. Solve the following.
- A) Explain the construction and working principle of a three-phase induction motor. 6
 - B) Explain V-curves of synchronous motors and their significance. 6
- Q4. Solve Any Two of the following.
- A) Describe the construction and working of a stepper motor. What are its types and applications? 6
 - B) Discuss the construction, working, and applications of a hysteresis motor. 6
 - C) Explain the working of a repulsion motor with a neat sketch. 6
- Q5. Solve Any Two of the following.
- A) Describe the working principle and applications of strain gauge. 6
 - B) Write a note on Hall Effect transducers and their use in magnetic field measurement. 6
 - C) Differentiate between analog and digital transducers with suitable examples. 6
- Q6. Solve Any Two of the following.
- A) What is electrical telemetry? Discuss its types and applications. 6
 - B) Explain the principle and working of computerized tomography. 6
 - C) Discuss the working and applications of an X-Y plotter. 6

*** End ***